EZEQUIEL ENOC JARAMILLO ANDRADE

AI, Computer Vision & Data Science Engineer | End-to-End ML Solutions for Startups & R&D

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EXECUTIVE SUMMARY & FREELANCE VALUE PROPOSITION

Applied AI, Computer Vision, and Data Science Engineer (Licensed Engineer & SENACYT M.Sc. Fellow in Cyber-Physical Systems) specializing in building production-ready data products and ML pipelines for tech startups and industrial clients. Proven track record architecting large-scale **Apache Parquet Medallion** pipelines, causal predictive models (**PyTorch, YOLOv8, Scikit-Learn**), interactive B2B web applications (**Streamlit, Plotly**), and **VNIR Hyperspectral** inspection systems. Co-author of peer-reviewed research in **MDPI Coatings (2025)** and certified by **MIT xPRO**, **OpenCV University**, and **IBM**.

TECHNICAL STACK & CORE COMPETENCIES

Machine Learning & Data Science: Python, Scikit-Learn, HistGradientBoosting, Out-of-Fold Target Encoding, Causal Inference, Statistical Modeling, MATLAB.

Data Engineering & Web Apps: Apache Parquet, Medallion ETL Architecture, Pandas, NumPy/SciPy (BLAS Vectorization), SQL, Streamlit, Plotly, Git/GitHub.

Computer Vision & Deep Learning: PyTorch, YOLOv8 (Ultralytics), OpenCV DNN (SSD ResNet), dlib, OpenFace (128-D), 2D FFT, VNIR Hyperspectral Imaging.

Cyber-Physical Systems, Drones & IoT: Licensed Drone Pilot (RPA Aerial Inspection), Optical Sensor Calibration, 3D CAD, CNC PCB Prototyping, Energy Analytics.

PRODUCTION AI & DATA ENGINEERING PORTFOLIO (GITHUB)

FlightOps Analytics — Medallion ETL Platform & Causal Air Traffic Disruption Predictor 6.57M Flights ROC-AUC 0.7898 github.com/EzequielJaramillo/flightops-air-traffic-analytics

- Engineered a 3-tier Medallion ETL pipeline (Bronze, Silver, Gold in **Apache Parquet**) across 6,573,267 US commercial flights and 130,305 daily weather records, cutting memory footprint by **>80%**.
- Trained a zero-leakage causal classifier (`HistGradientBoosting`, 70 pre-flight features) achieving **85.2% precision** in high-confidence regime, deployed via a 5-module interactive **Streamlit + Plotly** web platform.

YOLOv8 Real-Time Traffic Light Detection & State Classification mAP@0.5: 99.21% PyTorch GPU github.com/EzequielJaramillo/yolov8-traffic-light-detection

- End-to-end GPU training and video inference pipeline (`YOLOv8n/s`) achieving **mAP@0.5 = 0.9921**, **mAP@0.5:0.95 = 0.7913**, and **F1 = 0.96**, packaged with an interactive Streamlit dashboard.

Hyperspectral Euclidean Spectral Classifier (224 VNIR Bands) 83x BLAS Speedup 85.75% OA github.com/EzequielJaramillo/hyperspectral-euclidean-spectral-classifier

- Modular Python package for supervised hyperspectral cube classification using matrix-expanded Euclidean distance (BLAS), delivering an **83x speedup** (`34 ms` vs `2.83 s` over 78,000 pixels).

DoppelGänger Face Recognition Engine & Digital Image Processing Suite OpenFace 128-D 2D FFT github.com/EzequielJaramillo/doppelganger-celebrity-face-matcher

- Two-stage biometric pipeline (`OpenCV SSD ResNet-10` + 68-landmark affine alignment + `128-D` metric learning) plus 3 specialized repositories covering 2D FFT Notch filtering (**4.15x periodic noise suppression**), `RGBW/CMYKW` segmentation, and `PSNR/MSE` quantization.

PROFESSIONAL EXPERIENCE

Computer Vision, Deep Learning & Hyperspectral Researcher | GITTS Research Group (UTP) **Jan 2023 – Present**

- Architecting **Deep Learning** and spectral unmixing models for automated Non-Destructive Testing (NDT) and corrosion quantification on structural steel using VNIR hyperspectral imaging.
- Hardware-software integration: radiometric calibration of optical sensors, custom 3D CAD enclosures, and CNC PCB prototyping.

Data Applications & Energy Analytics Developer (Engineering Intern) | KilowattIA S.A. **Aug 2025 – Sep 2025**

- Developed and deployed a **web application** to digitize electrical load census forms across commercial and industrial facilities, streamlining field data collection.
- Built energy consumption analytics pipelines and executive visualizations to identify cost-saving opportunities and load optimization strategies for B2B clients.

Laboratory Instructor (Control Systems, Instrumentation & Circuits) | Technological University of Panama **Mar 2025 – Dec 2025**

- Led hands-on instruction and technical mentoring in automatic control systems, instrumentation, and experimental data analysis.

EDUCATION & PEER-REVIEWED PUBLICATIONS

M.Sc. in Energy Sustainability & Cyber-Physical Systems **In Progress**
Technological University of Panama — SENACYT Research Fellow

B.Sc. in Electrical Engineering **Graduated: 2025**
Technological University of Panama (UTP)

Peer-Reviewed Journal Publication (MDPI Coatings) **2025**
Co-author: *"Automated Assessment of Marine Steel Corrosion Using Visible–Near-Infrared Hyperspectral Imaging"*. doi.org/10.3390/coatings15060645

CERTIFICATIONS & LICENSES

- **MIT xPRO:** Artificial Intelligence Applied to Engineering and Science.
- **OpenCV University:** Advanced Vision Applications with Deep Learning & Transformers.
- **IBM Data Science Professional Certificate:** Machine Learning, SQL, Data Analysis & Python for AI.
- **Licensed Drone Pilot (Light RPA):** Civil Aeronautics Authority (AAC) — Aerial inspection.
- **Licensed Professional Engineer (JTIA):** ID 2026-013-001.
- **IEEE Leadership:** Vice-Chair IEEE-RAS (Robotics) & Secretary IEEE-ComSoc. Languages: Spanish (Native), English (Intermediate-Professional).